

Define - Graph, Epigraph, Convex Set, Convex Function.

Graph

The graph of a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, is the set of points in $\Omega \times \mathbb{R} \subset \mathbb{R}^{n+1}$ given by

$$\{[x, f(x)]^T : x \in \Omega\}$$

Epigraph

The epigraph of a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, denoted by $\text{epi}(f)$, is the set of points in $\Omega \times \mathbb{R}$ given by

$$\text{epi}(f) = \{[x, \beta]^T : x \in \Omega, \beta \in \mathbb{R}, \beta \geq f(x)\}$$

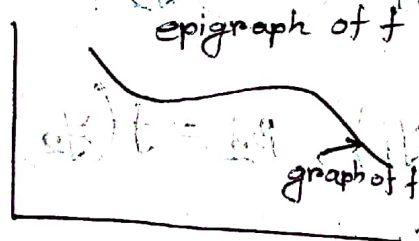


Fig: The graph and epigraph of a function $f: \mathbb{R} \rightarrow \mathbb{R}$.

Convex Set

A set $\Omega \subset \mathbb{R}^n$ is called convex if for every $x_1, x_2 \in \Omega$ and $\alpha \in (0, 1)$, $\alpha x_1 + (1-\alpha)x_2 \in \Omega$.

Convex Function

A function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is called convex on Ω if its epigraph is a convex set.

Theorem 21.1: - If a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is convex on Ω , then Ω is a convex set.

Proof: - We prove the theorem by contraposition. Suppose that Ω is not a convex set. Then there exist two points y_1 and y_2 such that for some $\alpha \in (0, 1)$,

$$z = \alpha y_1 + (1 - \alpha) y_2 \notin \Omega$$

Let

$$\beta_1 = f(y_1), \beta_2 = f(y_2)$$

Then the pairs

$$\begin{bmatrix} y_1 \\ \beta_1 \end{bmatrix}, \begin{bmatrix} y_2 \\ \beta_2 \end{bmatrix}$$

belong to the graph of f and hence also the epigraph of f .

$$\text{Let } w = \alpha \begin{bmatrix} y_1 \\ \beta_1 \end{bmatrix} + (1-\alpha) \begin{bmatrix} y_2 \\ \beta_2 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha y_1 + (1-\alpha) y_2 \\ \alpha \beta_1 + (1-\alpha) \beta_2 \end{bmatrix}$$

$$= \begin{bmatrix} z \\ \alpha \beta_1 + (1-\alpha) \beta_2 \end{bmatrix}$$

But since $z \notin \Omega$, $w \notin \text{epi}(f)$. Therefore, $\text{epi}(f)$ is not convex and hence f is not a convex function.

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Problem:- Let $f(x) = x_1 x_2$. Is f convex over

$$\Omega = \{x: x_1 \geq 0, x_2 \geq 0\}?$$

Solution:- Given that

$$f(x) = x_1 x_2$$

$$\text{and } \Omega = \{x: x_1 \geq 0, x_2 \geq 0\}$$

Let us consider $x = [1, 2]^T \in \Omega$ and $y = [2, 1]^T \in \Omega$

Then

$$\begin{aligned} 2x + (1-\alpha)y &= 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (1-\alpha) \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2-\alpha \\ 1+\alpha \end{bmatrix} \end{aligned}$$

Then

$$f(2x + (1-\alpha)y) = (2-\alpha)(1+\alpha) \\ = 2 + \alpha - \alpha^2$$

$$\text{and } \alpha f(x) + (1-\alpha)f(y) = 2$$

Let $\alpha = \frac{1}{2} \in (0, 1)$, then

$$f(2x + (1-\alpha)y) = 2 + \frac{1}{2} - \frac{1}{4} = \frac{9}{4}$$

Since

$$f(2x + (1-\alpha)y) > \alpha f(x) + (1-\alpha)f(y)$$

which shows that f is not convex over Ω .

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Theorem 21.3:- Let $f: \Omega \rightarrow \mathbb{R}$, $f \in C^1$, be defined on an open convex set $\Omega \subset \mathbb{R}^n$. Then f is convex on Ω if and only if for all $x, y \in \Omega$

$$f(y) \geq f(x) + Df(x)(y-x)$$

Proof:- Suppose $f: \Omega \rightarrow \mathbb{R}$ is differentiable and convex. Then for any $y, x \in \Omega$ and $\alpha \in (0, 1)$ we have

$$f(\alpha y + (1-\alpha)x) \leq \alpha f(y) + (1-\alpha)f(x)$$

$$\Rightarrow \alpha f(y) + (1-\alpha)f(x) \leq \alpha f(y) + f(x) - \alpha f(x)$$

$$\Rightarrow \alpha f(y) + f(x) - \alpha f(x) \leq f(x) + \alpha(f(y) - f(x))$$

$$\Rightarrow f(x) + \alpha f(y - x) \leq f(x) + \alpha(f(y) - f(x))$$

$$\Rightarrow f(x + \alpha(y - x)) \leq f(x) + \alpha(f(y) - f(x))$$

$$\Rightarrow f(x + \alpha(y - x)) - f(x) \leq \alpha(f(y) - f(x))$$

$$\Rightarrow \frac{f(x + \alpha(y - x)) - f(x)}{\alpha} \leq f(y) - f(x)$$

Now taking the limit as $\alpha \rightarrow 0$ and by the definition of the directional derivative of f at x in the direction $y - x$, we get

$$Df(x)(y - x) \leq f(y) - f(x)$$

$$\Rightarrow f(y) \geq f(x) + Df(x)(y - x)$$

Now conversely assume that Ω is convex, $f: \Omega \rightarrow \mathbb{R}$ is differentiable and for all $x, y \in \Omega$

$$f(y) \geq f(x) + Df(x)(y - x)$$

Let $u, v \in \Omega$ and $\alpha \in (0, 1)$. Because Ω is convex

$$w = \alpha u + (1-\alpha)v \in \Omega$$

We also have

$$f(u) \geq f(w) + Df(w)(u-w) \quad (1)$$

and

$$f(v) \geq f(w) + Df(w)(v-w) \quad (2)$$

Multiplying (1) by α and (2) by $(1-\alpha)$ and then adding, we get

$$\alpha f(u) + (1-\alpha)f(v) \geq f(w) + Df(w)(\alpha u + (1-\alpha)v - w)$$

But

$$w = \alpha u + (1-\alpha)v$$

$w - w$

$= 0$

Hence

$$\alpha f(u) + (1-\alpha)f(v) \geq f(\alpha u + (1-\alpha)v)$$

Hence f is a convex function.

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Theorem 21.5:- Let $f: \Omega \rightarrow \mathbb{R}$ be a convex function defined on a convex set $\Omega \subset \mathbb{R}^n$. Then a point is a global minimizer of f over Ω if and only if it is a local minimizer of f .

Proof:- The first part of the theorem is obvious. We prove the second part by contraposition.

Suppose that x^* is not a global minimizer of f over Ω . Then for some $y \in \Omega$, we have $f(y) < f(x^*)$. By assumption, the function f is convex and hence for all $\alpha \in (0, 1)$,

$$f(\alpha y + (1-\alpha)x^*) \leq \alpha f(y) + (1-\alpha)f(x^*)$$

Because $f(y) < f(x^*)$, we have

$$\alpha f(y) + (1-\alpha)f(x^*) = \alpha(f(y) - f(x^*)) + f(x^*) < f(x^*)$$

Thus for all $\alpha \in (0, 1)$

$$f(\alpha y + (1-\alpha)x^*) < f(x^*)$$

Hence there exist points that are arbitrarily close to x^* and have lower objective function value. For example, the sequence $\{y_n\}$ of points given by

$$y_n = \frac{1}{n}y + \left(1 - \frac{1}{n}\right)x^*$$

converges to x^* and $f(y_n) < f(x^*)$. Hence x^* is not a local minimizer, which completes the proof.

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convex > 0

concave ≤ 0

Example 21.6: Determine whether the following functions are convex, concave or neither

1. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -8x^2$

2. $f: \mathbb{R}^3 \rightarrow \mathbb{R}, f(x) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2$

$+ x_1x_3 - 3x_1 - 2x_2 + 15$

3. $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x) = 2x_1x_2 - x_1^2 - x_2^2$

Solution:

1. Given that

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -8x^2$$

$$\text{Now } \nabla f(x) = (-16x)$$

The Hessian matrix of f is

$$F(x) = -16 < 0, \text{ for all } x \in \mathbb{R}.$$

Hence, f is concave over $\mathbb{R}$.

2. Given that

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$f(x) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$$

$$\text{Now } \nabla f(x) = \begin{bmatrix} 8x_1 + 6x_2 + x_3 - 3 \\ 6x_2 + 6x_1 - 2 \\ 10x_3 + x_1 \end{bmatrix}$$

The Hessian matrix of f is

$$F(x) = \begin{bmatrix} 8 & 6 & 1 \\ 6 & 6 & 0 \\ 1 & 0 & 10 \end{bmatrix}$$

The leading principal minors of $F(x)$ are

$$\Delta_1 = 8 > 0$$

$$\Delta_2 = \det \begin{bmatrix} 8 & 6 \\ 6 & 6 \end{bmatrix} = 12 > 0$$

$$\Delta_3 = \det F(x) = 114 > 0$$

Hence, $F(x)$ is positive definite for all $x \in \mathbb{R}^3$.
Therefore, f is a convex over $\mathbb{R}^3$.

Given that

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = 2x_1x_2 - x_1^2 - x_2^2$$

$$\text{Now } \nabla f(x) = \begin{bmatrix} 2x_2 - 2x_1 \\ 2x_1 - 2x_2 \end{bmatrix}$$

The Hessian of f is

$$F(x) = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = 0$$

which is negative semidefinite for all $x \in \mathbb{R}^n$.
Hence f is concave on $\mathbb{R}^n$.

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Quadratic Form

A quadratic form $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a function

$$f(x) = x^T Q x$$

where Q is an $n \times n$ real matrix.

Proposition 21.1:- A quadratic form $f: \Omega \rightarrow \mathbb{R}$,

$\Omega \subset \mathbb{R}^n$, given by $f(x) = x^T Q x$, $Q \in \mathbb{R}^{n \times n}$, $Q = Q^T$ is convex on Ω if and only if for all $x, y \in \Omega$

$$(x-y)^T Q (x-y) \geq 0.$$

Solution:- We know that a function is convex if and only if for every $\alpha \in (0, 1)$ and every $x, y \in \mathbb{R}^n$, we have

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$$

$$\Rightarrow \alpha f(x) + (1-\alpha)f(y) - f(\alpha x + (1-\alpha)y) \geq 0$$

Substituting $f(x) = x^T Q x$ into the left-hand side of the above equation,

$$\begin{aligned}
 & \alpha x^T Q x + (1-\alpha) y^T Q y - (\alpha x + (1-\alpha) y)^T Q (\alpha x + (1-\alpha) y) \\
 &= \alpha x^T Q x + y^T Q y - \alpha y^T Q y - \alpha^2 x^T Q x \\
 &\quad - (2\alpha - 2\alpha^2) x^T Q y - (1 - 2\alpha + \alpha^2) y^T Q y \\
 &= \alpha(1-\alpha) x^T Q x - 2\alpha(1-\alpha) x^T Q y + \alpha(1-\alpha) y^T Q y \\
 &= \alpha(1-\alpha) (x-y)^T Q (x-y)
 \end{aligned}$$

Therefore, f is convex if and only if

$$\begin{aligned}
 & \alpha(1-\alpha) (x-y)^T Q (x-y) \geq 0 \\
 & \Rightarrow (x-y)^T Q (x-y) \geq 0
 \end{aligned}$$

which proves the result.



Calculation:

$$\begin{aligned} & 2x^T Q x + (1-2)y^T Q y - (2x + (1-2)y)^T Q (2x + (1-2)y) \\ &= 2x^T Q x + y^T Q y - 2y^T Q y - [(2x^T + y^T - 2y^T) Q (2x + (1-2)y)] \\ &= 2x^T Q x + y^T Q y - 2y^T Q y - [2\tilde{x}^T Q x + 2x^T Q y - 2\tilde{x}^T Q y \\ &\quad + 2y^T Q x + y^T Q y - 2y^T Q y - 2\tilde{y}^T Q x \\ &\quad - 2y^T Q y + 2\tilde{y}^T Q y] \\ &= 2x^T Q x + y^T Q y - 2y^T Q y - [2\tilde{x}^T Q x + (2-2\tilde{x})x^T Q y \\ &\quad + (2-2\tilde{x})y^T Q x + (1-2\alpha+2\tilde{x})y^T Q y] \\ &= (2-2\tilde{x})x^T Q x + (2-2\tilde{x})y^T Q y - (2\alpha-2\tilde{x})x^T Q y \\ &= \alpha(1-\alpha)x^T Q x - 2\alpha(1-\alpha)x^T Q y + \alpha(1-\alpha)y^T Q y \\ &= \alpha(1-\alpha)(x^T Q x - 2x^T Q y + y^T Q y) \\ &= \alpha(1-\alpha)(x^T Q x - x^T Q y - x^T Q y + y^T Q y) \\ &= \alpha(1-\alpha)(x^T Q x - x^T Q y - y^T Q x + y^T Q y) \\ &= \alpha(1-\alpha)[x^T Q (x-y) - y^T Q (x-y)] \\ &= \alpha(1-\alpha)(x-y)^T Q (x-y) \end{aligned}$$